Bonus Challenge 4 — Building Agents with Google AI Applications
Aurora Bay Agent (US-Central1)

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1. Overview

This document explains, step by step, how I designed, configured, and validated a conversational agent for the fictional town of Aurora Bay, Alaska, using Google AI Applications. The agent uses a Playbook and a Vertex AI Data Store hooked to a Google Cloud Storage bucket so that answers are grounded in the town's FAQs rather than model guesses.

2. Goal and Requirements

Goal: Demonstrate the ability to build an AI Applications Conversational Agent that uses a Playbook and Data Store to answer user questions about Aurora Bay.

Requirements met:

- Created a Conversational Agent app named "Aurora Bay Agent" in region US-Central1.
- Configured a Vertex AI Data Store and connected it to cloud storage: gs://labs.roitraining.com/aurora-bay-faqs
- Authored a Default Playbook with a clear goal, instructions, and the linked data store as the grounding source.
- Tested the agent to confirm answers are based on the data store (and do not hallucinate).

3. High-Level Architecture

The solution follows a retrieval-augmented generation (RAG) pattern: user questions are routed through a Playbook that invokes a Data Store tool. The Data Store indexes the Aurora Bay FAQs in Cloud Storage and returns the most relevant passages. The agent then composes grounded responses, preferring cited content from the store and avoiding unsupported claims.

4. Step-by-Step Implementation

4.1 Create the Conversational Agent

1. Open Google Cloud Console → AI Applications → Agents.
2. Click Create Agent and name it Aurora Bay Agent. Choose region as us-central1.
3. Keep default runtime settings and create the app.

4.2 Configure the Default Playbook

I edited the Default Playbook as follows:

- **Goal:** "Answer frequently asked questions for citizens of Aurora Bay, Alaska, with accurate, grounded information."
- **Instructions:** "Be concise, cite facts only from the connected data store. If a question is outside the data, ask a clarifying question or state that the information is unavailable."

4.3 Add and Connect the Data Store

4. From the Playbook, add a Tool → Data stor
5. Create or select a Vertex AI Search Data Store and point it to gs://labs.roitraining.com/aurora-bay-faqs.
6. Enable passage-based retrieval with balanced precision.
7. Save the tool and ensure the Playbook uses it for knowledge-grounded answers.

4.4 Guardrails and Hallucination Mitigation

- Prompt discipline: Instructions explicitly require citing the data store as the sole knowledge source.
- Fallback behavior: When no relevant snippets are retrieved, the agent replies that the information is not available and invites the user to rephrase.
- Testing queries: I verified that off-topic questions led to safe, non-speculative responses.

4.5 Testing the Agent

- Asked "What services does Aurora Bay offer?" and verified the answer matched the FAQs.
- Explored capabilities by asking "What options do you have?" to confirm the agent lists topical areas (e.g., services, attractions, history).
- Checked multiple phrasings and ensured consistent, grounded responses.

5. Screenshots

image (6).png

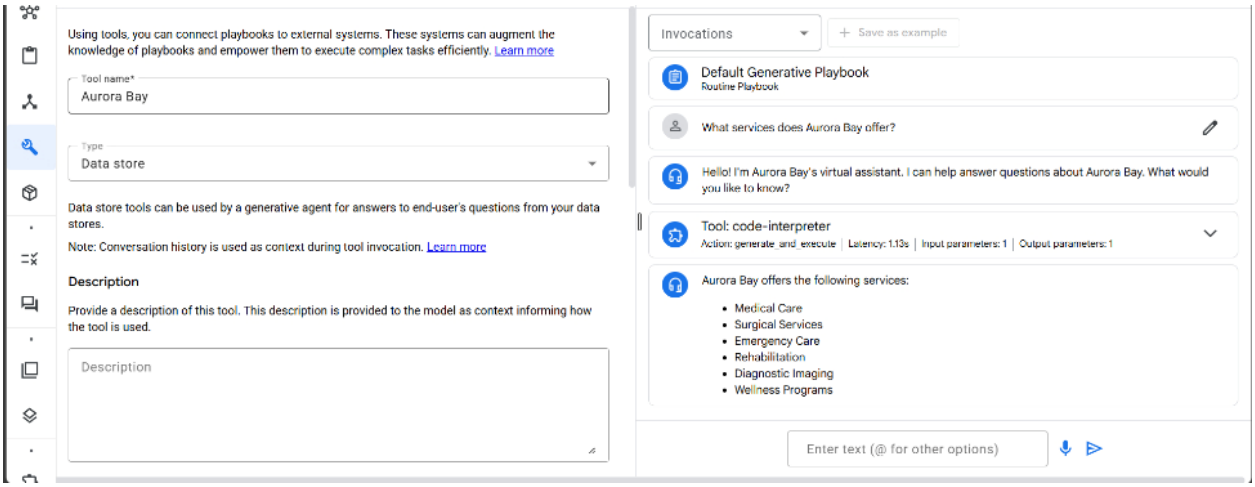

Figure: image (6)

Figure: image (5)

Appendix A — Key Config Values

- Agent name: Aurora Bay Agent
- Region: us-central1
- Data source: gs://labs.roitraining.com/aurora-bay-faqs
- Playbook: Default Generative Playbook with Data Store tool attached